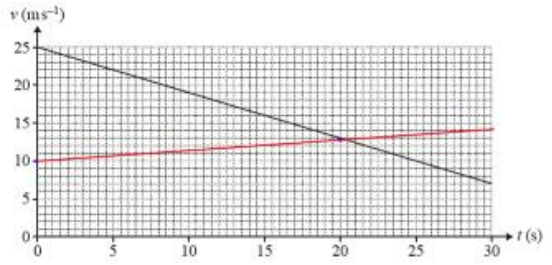
Question			Answer	Marks	AO	Guidance
9	(a)		The student has found the magnitude of the acceleration and not the actual acceleration which is negative. The argument is not valid.	B1 [1]	2.3	Must comment of the validity of the argument. Allow for a comment that student is wrong which includes statement that the acceleration should be negative or similar.
9	(b)		$v_B = 10 + 0.15t$	B1 [1]	3.3	Award if $v = 10 + 0.15t$ seen
9	(c)			B1 [1]	1.1	Line through coordinates (0,10) and (20,13) FT their linear (b)
9	(d)		Same speed when $25 - 0.6t = 10 + 0.15t$, giving $t = 20$	B1	3.4	Soi FT their (b)
			Displacement of train A is 380 m Displacement of train B is 230 m	M1	3.4	Attempt to use area or *suvat* equation(s) to find the displacement of at least one train at their value for t
			So distance between them is 150 m	A1	1.1	cao
			Alternative for last 2 marks			
			Area of the triangle between graphs is $\frac{1}{2} \times (25 - 10) \times 20$	M1		Soi FT their (b)
			So distance between them is 150 m	A1		cao

Question			Answer	Marks	AO	Guidance
				[3]		
9	**(e)**		[Velocity is zero at 41.7 s.] For large values of *t*, the model predicts that the velocity will be negative but the train will not go backwards as the trains are travelling in the same direction	**B1** **[1]**	**3.5b**	Must compare what happens in the model with what happens to the train. (See appendix for exemplars)

Exemplar responses for Q9(e)

Response	Mark
“The velocity might be 0 and it doesn’t stop”	**B0**
“The velocity would be negative” on its own	**B0**
“The velocity would be negative which is impossible”	**B0**
“The velocity would be negative and velocity can’t be negative”	**B0**
“Because the model would represent the train going backwards”	**B1**
“The velocity would be negative which means going backwards”	**B1**
“The velocity would be negative which means going backwards which it can’t do”	**B1**

“The velocity would be negative which means going backwards which it says it doesn’t do in this question”	**B1**
“It would give an ever increasing negative velocity”	**B0**
“The model predicts travelling infinitely fast which is not possible”	**B1**
“The model predicts travelling infinitely fast in the opposite direction which is not possible”	**B1**